A NOETHERIAN LOCAL RING WITH GEOMETRICALLY NORMAL FORMAL FIBRES IS NAGATA AND HAS THE SAME NUMBER OF BRANCHES AS ITS COMPLETION

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Allowed standard prerequisites (editorial).

- Nagata rings and finiteness of the integral closure in the fraction field. used at proof lines 32242-32243 and 32263-32264. Supplies both the Nagata conclusion and the finiteness that makes the branch count of A equal to the number of maximal ideals of A'.
- P-rings for a property of formal fibres, and the criterion on maximal ideals. invoked at proof lines 32244-32248. Propagates geometric normality of the formal fibres to the auxiliary rings used in the reduction.
- Normality of the completion of a local ring with geometrically normal formal fibres. invoked at proof lines 32272-32275. Is what makes the minimal primes of A' tensor_A A-hat match the factors of the product decomposition.
- Flatness of the completion map and commutation of completion with finite extensions. used at proof lines 32268-32271 and 32278-32281. Gives the decomposition of A'-hat and the chain of inclusions whose minimal primes are compared.

Source reference. Original lemma 07N9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9754–9754. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

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Source reference. Original lemma 05WQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7594–7594. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 13413: sources/stacks-more-algebra/pinned-original-source.tex, source lines 13413–13413. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 31155: sources/stacks-more-algebra/pinned-original-source.tex, source lines 31155–31155. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 12954: sources/stacks-more-algebra/pinned-original-source.tex, source lines 12954–12954. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 13485: sources/stacks-more-algebra/pinned-original-source.tex, source lines 13485–13485. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 13215: sources/stacks-more-algebra/pinned-original-source.tex, source lines 13215–13215. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 13155: sources/stacks-more-algebra/pinned-original-source.tex, source lines 13155–13155. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 10979: sources/stacks-more-algebra/pinned-original-source.tex, source lines 10979–10979. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original proposition 12999: sources/stacks-more-algebra/pinned-original-source.tex, source lines 12999–12999. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $(A, \mathfrak{m})$ be a Noetherian local ring. If the formal fibres of A are geometrically normal (for example if A is excellent or quasi-excellent), then A is Nagata and the number of (geometric) branches of A and $A^\wedge$ is the same.*

Proof. Since a normal ring is reduced, we see that A is Nagata by Lemma 13413. In the rest of the proof we will use Lemma 13215, Proposition 12999, and Lemma 12954. This tells us that A is a P-ring where $P(k \to R) =$"R is geometrically normal over k" and the same is true for any (essentially of) finite type A-algebra.

Let $\mathfrak{q} \subset A$ be a minimal prime. Then $A^\wedge/\mathfrak{q}A^\wedge = (A/\mathfrak{q})^\wedge$ and $A^h/\mathfrak{q}A^h = (A/\mathfrak{q})^h$ (Algebra, Lemma 05WQ). Hence the number of branches of A is the sum of the number of branches of the rings $A/\mathfrak{q}$ and similarly for $A^\wedge$. In this way we reduce to the case that A is a domain.

Assume A is a domain. Let A' be the integral closure of A in the fraction field K of A. Since A is Nagata, we see that $A \to A'$ is finite. Recall that the number of branches of A is the number of maximal ideals $\mathfrak{m}'$ of A' (Lemma 31155). Also, recall that

$$(A')^\wedge = A' \otimes_A A^\wedge = \prod\nolimits_{\mathfrak{m}' \subset A'} (A'_{\mathfrak{m}'})^\wedge$$

by Algebra, Lemma 07N9. Because $A'_{\mathfrak{m}'}$ is a local ring whose formal fibres are geometrically normal, we see that $(A'_{\mathfrak{m}'})^\wedge$ is normal (Lemma 13485). Hence the minimal primes of $A' \otimes_A A^\wedge$ are in 1-to-1 correspondence with the factors in the decomposition above. By flatness of $A \to A^\wedge$ we have

$$A^\wedge \subset A' \otimes_A A^\wedge \subset K \otimes_A A^\wedge$$

Since the left and the right ring have the same set of minimal primes, the same is true for the ring in the middle (small detail omitted) and this finishes the proof.

To see this is true for the number of geometric branches we use the result for branches, the fact that the formal fibres of A^{sh} are geometrically normal (Lemmas 13215 and 13155) and the fact that $(A^{sh})^\wedge = ((A^\wedge)^{sh})^\wedge$ by Lemma 10979. $\square$